

and scriptures have been talking about quantum, and it is the basis of the Indian Vedanta system. So, we want to build it here and take it to the world.” Thanks to QNu Labs, India might soon become the go-to place for quantum cybersecurity.

QNu Labs has partners all across the globe. In India, they have partnered with big players like BEL (Bharat Electronics Limited) and provide services to govern-

ment and defense organisations. The company was a part of several incubation platforms like Cisco Launchpad, Intel Startup Program, Nasscom Deep Tech Club and of course, IIT Madras, where it was founded.

It all started in 2016, when India had just started becoming a digital economy. “We felt that if a billion people have to use digital platforms on a daily basis, then the

security needs to be very different,” says Gupta. The existing security systems are about 30 to 40 years old and have already been exposed. They’re vulnerable to algorithms. “We’re helping enterprises adopt quantum tech today so that in the future, technology equipped with quantum computing cannot compromise our data and systems,” adds Gupta.

—Aaryaa Padhyegurjar

2 MAKING METAL 3D PRINTERS MORE ACCESSIBLE

Imagine being able to print your metal parts at your place without requiring any expensive machining tools. A Chile based inventor, Agustin Cruz, has developed a working prototype of a low-cost metal 3D printer, thus accelerating innovations and reducing barriers to scientific research

To democratise science and accelerate research and innovations in the lower-income societies, Agustin, an electrical engineer with over ten years of experience in building laser machines, robots, laboratory equipment, and many other exciting tools, decided to build a 3D metal printer. “3D metal printing is an impactful tool and making it more accessible will democratise science and engineering and empower people. This low-cost hardware will be able to accelerate innovations and lower barriers to scientific research,” says Agustin, when asked about his motives behind investing time and resources in this project.

3D printers that use a plastic filament are common these days. They are cost-effective and convenient for rapid prototyping, especially during the proof-of-concept phase. But when asked what advantages 3D metal printers offer over regular printers, Agustin says, “Regular 3D printers are used to make plastic objects, which are good for prototypes



Agustin Cruz, the innovator of metal 3D printer

and specific applications, whereas 3D metal printers work with titanium powder, stainless steel powder, aluminium alloys, and other alloys. So, if you need a durable, strong part that is resistant to high temperatures, regular printers are definitely not the

best choice.”

A metal 3D printer is a machine capable of making intricate metal parts. Unlike the other metal fabrication processes, like CNC and lathing, which are based on subtractive manufacturing, metal 3D printing is